

# Trapped in a Low Dimension:

## Decoding Visual Stimulus Identity from V1 Population Activity

Omar Jallouq   Filip Janczak   Mohammed Aaaqib Khatibi   Dhruv Singh

### 1 Introduction

A central question in systems neuroscience is whether and how the activity of populations of neurons in primary visual cortex (V1) encodes the identity of visual stimuli. V1 is the first cortical relay of retinal input, and modern two-photon calcium imaging — exemplified by the MICrONS dataset [1] — makes it possible to record hundreds of genetically identified, anatomically reconstructed neurons simultaneously. This setting directly motivates the population-coding and decoding-theory frameworks covered in class, and lends itself to dimensionality-reduction analysis (PCA, t-SNE, UMAP) and modern contrastive-learning embeddings such as CEBRA [2].

We exploit MICrONS session 9\_4 to ask whether the coarse *category* of a visual stimulus (natural movie clip, Monet2 noise, Trippy grating) can be read out from the response of  $\sim 579$  proofread V1 excitatory neurons, whether retaining the within-trial temporal structure improves decoding, and whether the primary thalamo-cortical input layer L4 carries a sharper category signal than the associative layer L2/3 or the full population.

### 2 Research Questions

**Q1.** Can stimulus category be decoded from the trial-averaged mean-rate population response of V1 above a leakage-free shuffle baseline?

**Q2.** Does retaining within-trial temporal structure (per-bin samples) improve decoding performance and embedding quality?

**Q3.** Does L4 carry a stronger category signal than L2/3, and does any single layer outperform the full-population decoder?

### 3 Methodology

**Dataset & filtering.** MICrONS v1718, session 9\_4. Neurons restricted to V1, proofread axons (`ax_clean`), matched functional tuning, excitatory only: **579 trusted neurons** (L1:1, L2/3:220, L4:257, L5:89, L6:12) across **280 unique stimuli** in **464 trials** (240 Clip, 20 Monet2, 20 Trippy hashes). Preprocessing: drop neurons with  $>10\%$  NaN, mean-impute the rest, drop constant stimuli, `StandardScaler`, class-balanced subsample.

**Two-phase pipeline.** *Part I — Trial-averaged.* Each unique stimulus hash collapses to a single  $1 \times 579$  mean-response vector across repetitions and frames ( $\sim 280$  samples). PCA, t-SNE (perp 15/30/50), UMAP ( $nn$  5/15/50) and CEBRA are computed; a logistic regression is evaluated on the 2D embeddings under `StratifiedKfold(5)` on the balanced subsample. The CV is leakage-free: scaler, PCA, UMAP and CEBRA are wrapped in an `sklearn Pipeline` and re-fitted on each training fold. Significance is assessed by 100 stimulus-level label permutations.

*Part II — Per-bin temporal.* Each trial is segmented into non-overlapping 5-frame bins, producing  $\sim 7,520$  samples

( $bin \times 579$ ). Decoding is evaluated in the full PCA-50, UMAP-2D and CEBRA-8D spaces using `StratifiedGroupKfold(5)` grouped by stimulus hash, so the same stimulus identity never co-occurs in train and test. All transforms are re-fitted inside each fold; CEBRA receives bins sorted by (`trial`, `time`) as required by its offset-10 receptive field. The null distribution is built by trial-level label permutations, which preserve the within-trial temporal autocorrelation. The pipeline is repeated, fold-locally, restricted to L2/3, L4, L5 and L6 individually. Code and seeds (`SEED=42`) are fully reproducible.

### 4 Results

**Part I.** On the trial-averaged data all linear/topological embeddings are significantly above the 0.333 chance level (Table 1); UMAP ( $nn=15$ ) gives the strongest 2D separation (bal. acc. 0.575). The PCA scree shows V1 activity lives in a *high-dimensional* space: 82 PCs are needed for 80% variance and the participation ratio is 47.2, ruling out a tidy low-rank representation. CEBRA reports a higher 2D score (0.765), but with only  $\sim 100$  balanced samples and no temporal context this is suspect — motivating Part II.

**Table 1:** Part I — trial-averaged decoding (2D embeddings, `StratifiedKfold`). Chance 0.333. \*\*\*  $p < 0.001$  vs. 100-shuffle null.

| Method           | Bal. Acc. | Macro F1 |
|------------------|-----------|----------|
| PCA-2D           | 0.533***  | 0.402*** |
| t-SNE ( $p=30$ ) | 0.401***  | 0.339*** |
| UMAP ( $nn=15$ ) | 0.575***  | 0.389*** |
| CEBRA-2D         | 0.765***  | 0.739*** |

**Part II.** Retaining temporal structure produces a  $>70\times$  increase in samples and a marked improvement in decoding (Table 2, Fig. 3). PCA-50 reaches  **$0.822 \pm 0.06$**  balanced accuracy, far above the trial-level shuffle null ( $0.334 \pm 0.03$ ,  $p < 0.001$ , Fig. 4). CEBRA-8D, which can finally exploit a temporal signal, achieves  **$0.899 \pm 0.03$**  bal. acc. (F1 0.867). UMAP-2D, by contrast, *drops* to 0.387 — a leakage-free 2D embedding is not expressive enough for the per-bin task. The confusion matrices show that CEBRA recovers the minority Monet2 and Trippy classes (recall 0.82 and 0.86) that PCA-50 still partially confuses.

**Table 2:** Part II — leakage-free `StratifiedGroupKfold` on  $\sim 7,520$  per-bin samples. Chance 0.333; majority baseline (always Clip) 0.766.

| Method          | Bal. Acc.                          | Macro F1                           |
|-----------------|------------------------------------|------------------------------------|
| PCA-2D          | $0.481 \pm 0.06$                   | $0.364 \pm 0.05$                   |
| <b>PCA-50D</b>  | <b><math>0.822 \pm 0.06</math></b> | <b><math>0.714 \pm 0.05</math></b> |
| UMAP-2D         | $0.387 \pm 0.05$                   | $0.331 \pm 0.04$                   |
| <b>CEBRA-8D</b> | <b><math>0.899 \pm 0.03</math></b> | <b><math>0.867 \pm 0.02</math></b> |

**Layer-specific decoding (Part II).** The full popula-

tion dominates every individual layer (Table 3, Fig. 5). L4 (257 neurons) edges L2/3 (220) very narrowly — 0.736 vs. 0.730 for PCA-50, and 0.784 vs. 0.782 for CEBRA-8D — well within standard deviation. The smaller L5 (89) and L6 (12) layers fall sharply behind.

**Table 3:** Per-layer Part II decoding (bal. acc.,  $\pm$  s.d. across folds).

| Layer      | $n$        | PCA-50                             | CEBRA-8D                           |
|------------|------------|------------------------------------|------------------------------------|
| L2/3       | 220        | $0.730 \pm 0.05$                   | $0.782 \pm 0.04$                   |
| L4         | 257        | $0.736 \pm 0.05$                   | $0.784 \pm 0.09$                   |
| L5         | 89         | $0.586 \pm 0.05$                   | $0.576 \pm 0.07$                   |
| L6         | 12         | $0.506 \pm 0.03$                   | $0.520 \pm 0.04$                   |
| <b>All</b> | <b>579</b> | <b><math>0.822 \pm 0.06</math></b> | <b><math>0.899 \pm 0.03</math></b> |

## 5 Discussion

**Q1 — Yes, with reservation.** Coarse stimulus category is decodable from trial-averaged V1 mean rates significantly above shuffle null, but the modest effect sizes (bal. acc. 0.53–0.58 for PCA/UMAP) make this a weak readout in absolute terms. The high-dimensional structure of V1 (47-dim participation ratio) means  $k = 2$  embeddings discard most of the relevant variance.

**Q2 — Strongly yes.** Per-bin temporal binning raises PCA decoding from 0.53 to **0.82** and unlocks CEBRA, whose contrastive objective explicitly exploits temporal contiguity, to **0.90**. The qualitative CEBRA embedding (Fig. 4) shows clean category clusters that the Part I 2D plots (Fig. 2) lack. Beyond the methodological win, this is a substantive finding: *when* a stimulus is presented, not just *which* neurons respond, is informative.

**Q3 — Layer differences are real but small, and the full population always wins.** The clean hierarchical  $L4 > L2/3$  picture suggested by the Part I results does not survive the more rigorous Part II protocol: L4 only marginally outperforms L2/3 ( $\Delta \approx 0.002$  on CEBRA,  $\Delta \approx 0.006$  on PCA-50) — within fold s.d. The small- $n$  layers (L5, L6) clearly underperform, but a fair test would equalise neuron counts. The robust take-away is that **category information is distributed across layers**, and pooling them yields a clear  $\sim 10$ – $15$  pt gain over any single layer. This is consistent with the view that L4 carries the sharpest *drive* but downstream layers contribute complementary signal that a linear decoder can use.

**Limitations.** (i) *Class imbalance:* 240/280 stimuli are Clips; even balanced subsampling cannot fully correct for it, and the per-bin majority baseline is 0.766 — only the PCA-50 and CEBRA scores meaningfully exceed it. (ii) *Calcium imaging temporal resolution:* GCaMP6s has  $\sim 300$  ms decay, so 5-frame bins partially average over the underlying spiking dynamics; finer bins might help or hurt depending on signal-to-noise. (iii) *Linear classifier only:* a logistic regression cannot exploit non-linear manifold structure that may exist beyond what UMAP/CEBRA already linearise. (iv) *Single session, one mouse:* generalisation across animals is untested. (v) *Unequal layer  $n$ :* the  $L4 \approx L2/3$  comparison is fair, but L5/L6 are confounded by sample size.

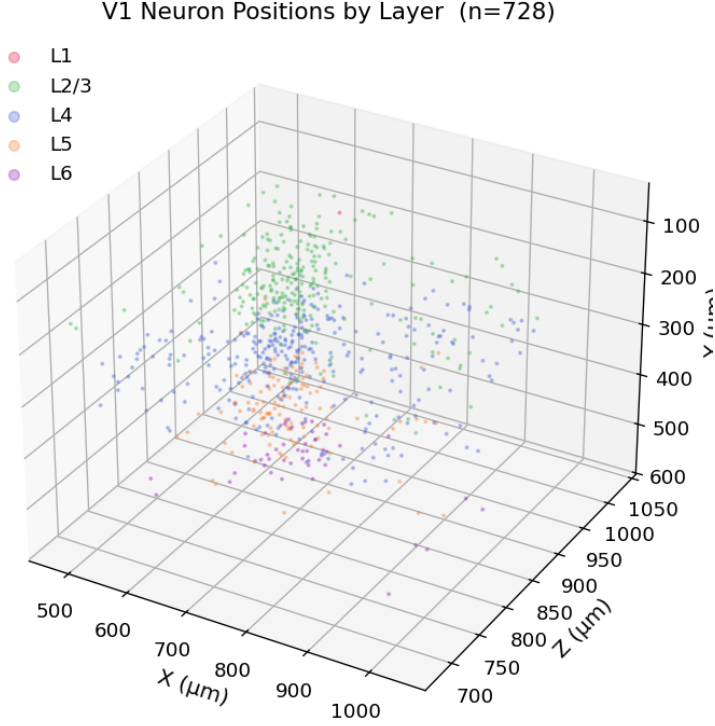
**Future work.** Decode *within-category* identity (specific

clip ID or grating orientation) to test whether V1 carries fine information beyond category; sweep bin sizes (1, 2, 5, 10 frames) for the optimal temporal resolution; equalise per-layer neuron counts via random subsampling; replace logistic regression with an SVM-RBF or shallow MLP.

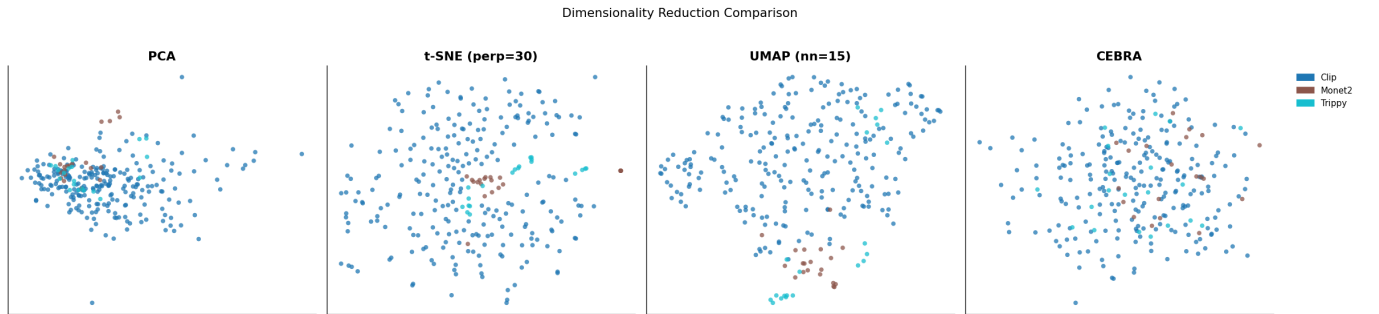
## References

- [1] MICrONS Consortium. Functional connectomics spanning multiple areas of mouse visual cortex. *Nature* 640, 435–447 (2025).
- [2] Schneider, S., Lee, J.H., Mathis, M.W. Learnable latent embeddings for joint behavioural and neural analysis. *Nature* 617, 360–368 (2023).
- [3] McInnes, L., Healy, J., Melville, J. UMAP: Uniform Manifold Approximation and Projection. *arXiv:1802.03426* (2018).
- [4] van der Maaten, L., Hinton, G. Visualising data using t-SNE. *JMLR* 9, 2579–2605 (2008).

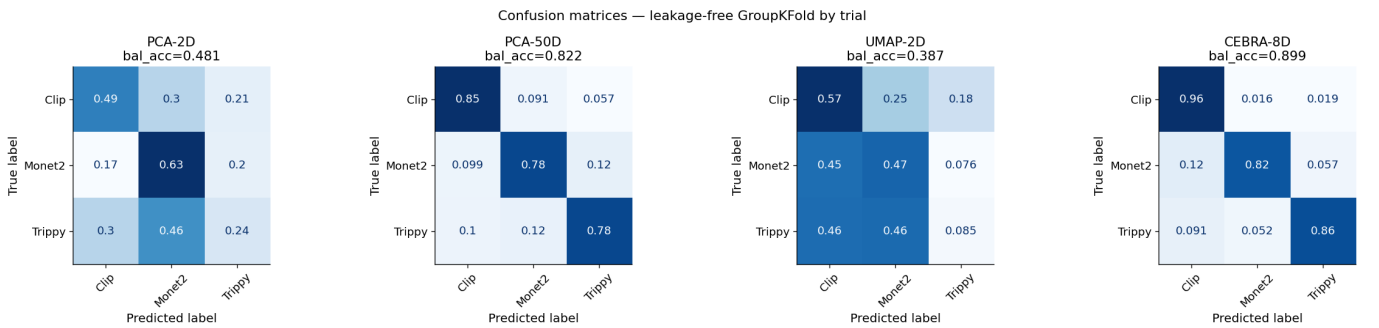
## Appendix — Supplementary Figures



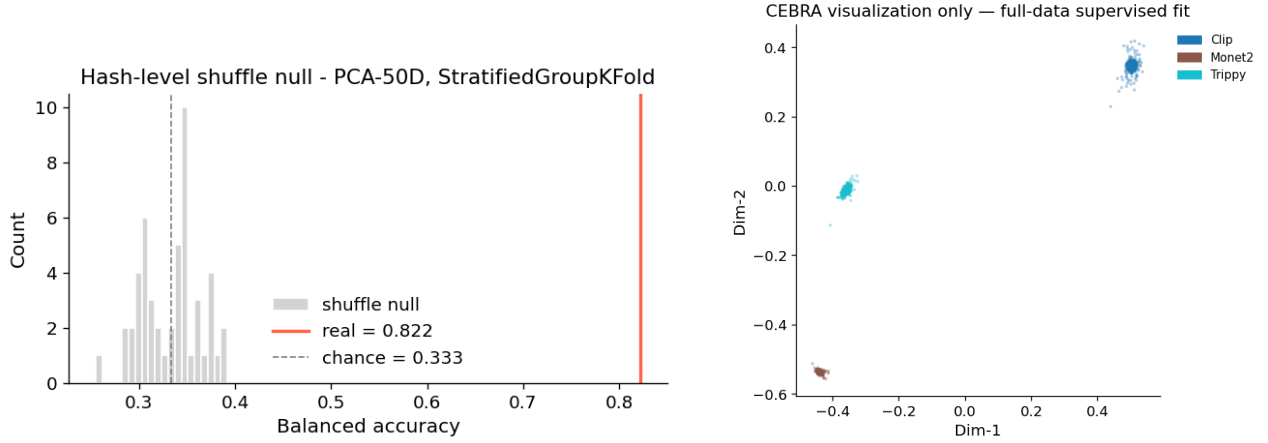
**Figure 1:** Anatomical 3D positions of the 728 V1 proofread excitatory neurons in session 9\_4, coloured by cortical layer (L1–L6). MICrONS v1718.



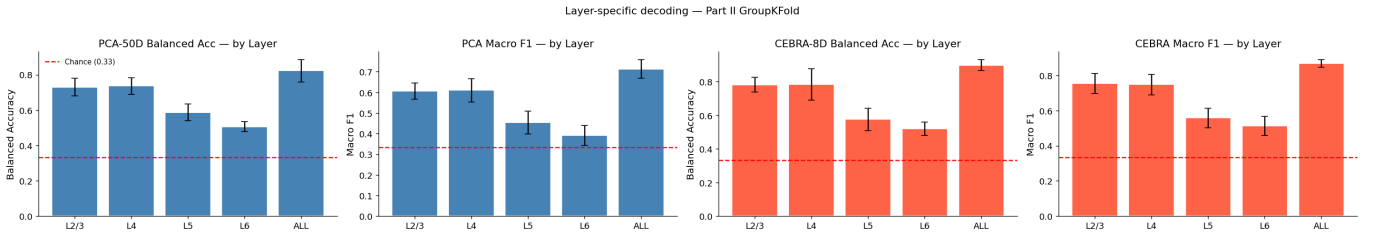
**Figure 2: Part I — 2D embeddings on the 280 trial-averaged stimuli.** Colours: Clip (blue), Monet2 (brown), Trippy (cyan). PCA and UMAP show partial category separation; t-SNE is noisier; CEBRA is shown for visualisation but is unreliable in Part I because it has no temporal context.



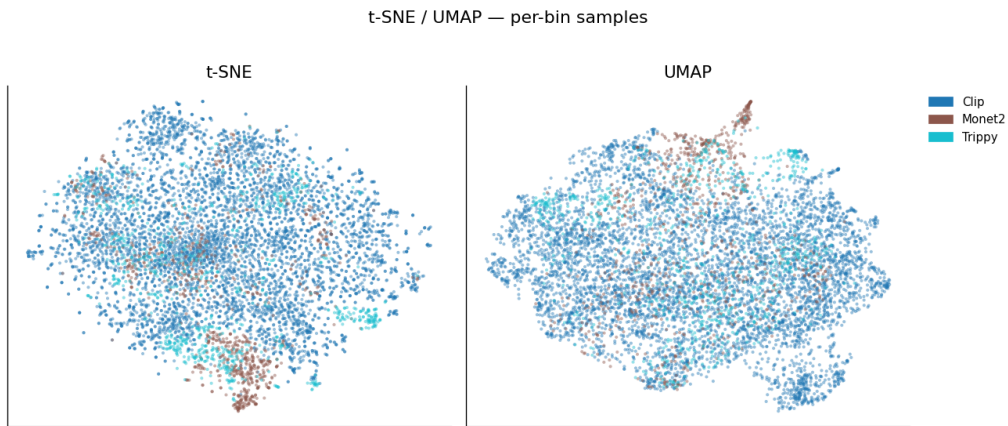
**Figure 3: Part II — confusion matrices, leakage-free StratifiedGroupKFold by stimulus hash.** PCA-50 (bal. acc. 0.822) and CEBRA-8D (0.899) recover the minority Monet2/Trippy classes that PCA-2D and UMAP-2D miss.



**Figure 4:** **Left:** Part II trial-level shuffle null for PCA-50 (mean  $0.334 \pm 0.03$ , max 0.392) vs. real score (0.822); empirical  $p < 0.001$ . **Right:** CEBRA-8D embedding (Part II, full-data supervised fit, visualisation only) showing clean per-category clusters — referenced in Q2 of the discussion.



**Figure 5: Per-layer Part II decoding** (PCA-50 left two panels, CEBRA-8D right two; balanced accuracy and macro F1). L2/3 and L4 are essentially tied, both clearly below the full-population (ALL) decoder. Error bars: s.d. across folds. Red dashed line: chance.



**Figure 6: Part II — t-SNE / UMAP embeddings on per-bin samples** (8,000-bin subsample). The Clip class dominates by sample count; Monet2 and Trippy form weak, partially overlapping pockets. Compare to the Part II CEBRA embedding above, which separates the three categories cleanly.